

HYPERTENSION CARDIOLOGY DATA CHALLENGE

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Predicting BP Control Failure and Evaluating Drug Outcomes across Comorbidity Profiles.

PROJECT BACKGROUND

This project was completed as part of the Zion Tech Hub Data Challenge, which focuses on a Hypertension Cardiology Center dataset containing 350 patient records. The goal was to solve a real problem by analyzing patient outcomes, identifying risk factors for poor blood pressure control and recommending better treatment strategies.

The challenge focuses on two major questions?

- What predicts BP control failure at 3 months, and can we identify high-risk patients at the point of prescription?
- Which antihypertensive drug class delivers the best outcomes for patients with specific comorbidity profiles?

Tools Used:

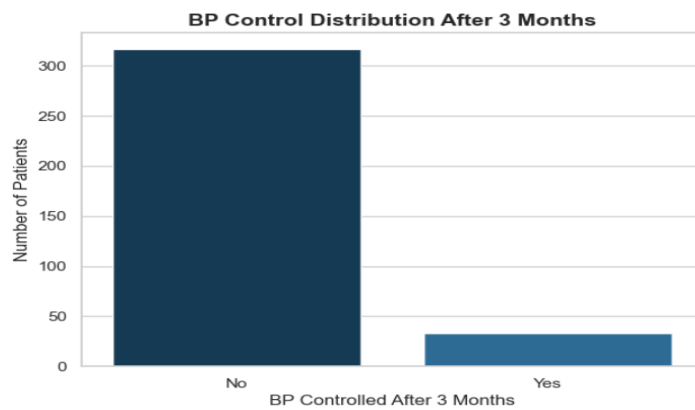
- Excel: Data Cleaning and Exploration.
- PostgreSQL: Granular analysis and Insight Generation.
- Python(Jupyter Notebook): Predictive Modeling, high-risk patient identification and data visualization.

DATA EXPLORATION IN EXCEL

- Standardized data types.
- Grouped patient age into 6 categories.
- Created two new columns for BP systolic and diastolic reduction.

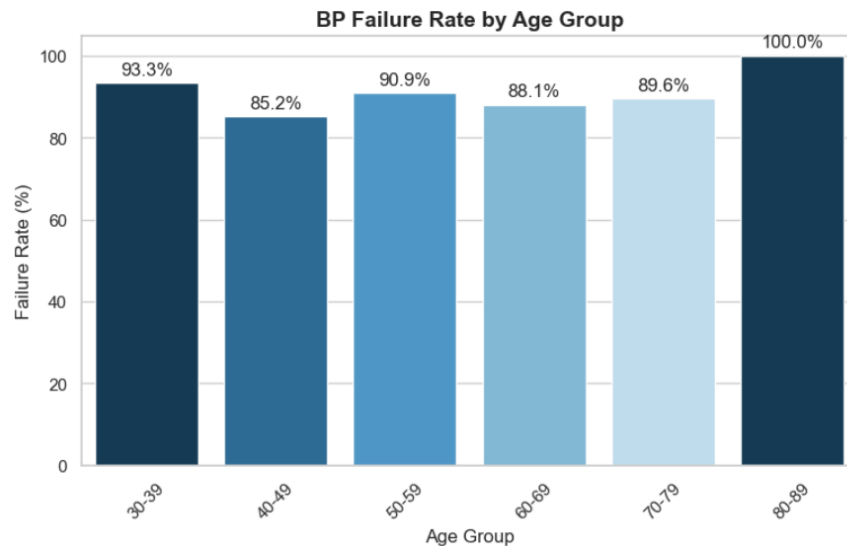
INSIGHTS FROM THE IN-DEPTH ANALYSIS IN SQL

1. Overall BP Control Performance



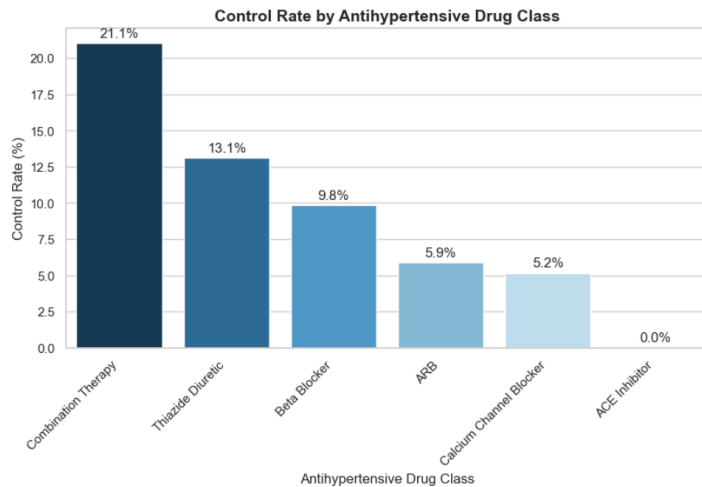
Out of 350 patients, 317 patients have their BP not controlled after 3 months, making it necessary to identify the factors responsible for poor outcomes.

2. Age Group and BP Control Outcome



Some age groups showed higher BP control after 3 months in this order, 80-89(100%), 30-39(93.33%), 50-59(90.91%), 70-79(89.55%), 60-69(88.14%) and 40-49(85.25%) although age alone should not be treated as the only driving factor.

3. Drug class overall performance



Drug effectiveness varies across the antihypertensive class in this order Combination Therapy(21.05%), Thiazide Diuretic(13.11%), Beta blocker(9.84%), ARB(5.88%), Calcium Channel Blocker(5.17%) and Ace inhibitor(0%). This performance confirmed that treatment outcome is not only about prescribing medication but also about matching the right drugs to the right patient.

- 4. Gender and BP Outcome:** Females has a higher control rate of 11.24% when compared to Males(7.56%).
- 5. Gender and Medical Adherence:** The male gender has an overall poor adherence resulting to a higher failure rate when compared to females.
- 6. Gender and Drug class:** The effectiveness of drugs for males and females are in this order Combination Therapy, Thiazide Diuretic, Beta Blocker and ARB but the control rate for all drugs proved to be more effective in females than in males which can be due to more medical adherence from the females.
- 7. Comorbidity Impact:** Patient with comorbidities had higher BP failure rate in comparison with those without. This means that patients with existing chronic conditions require closer monitoring and possible more personalized treatment plans.

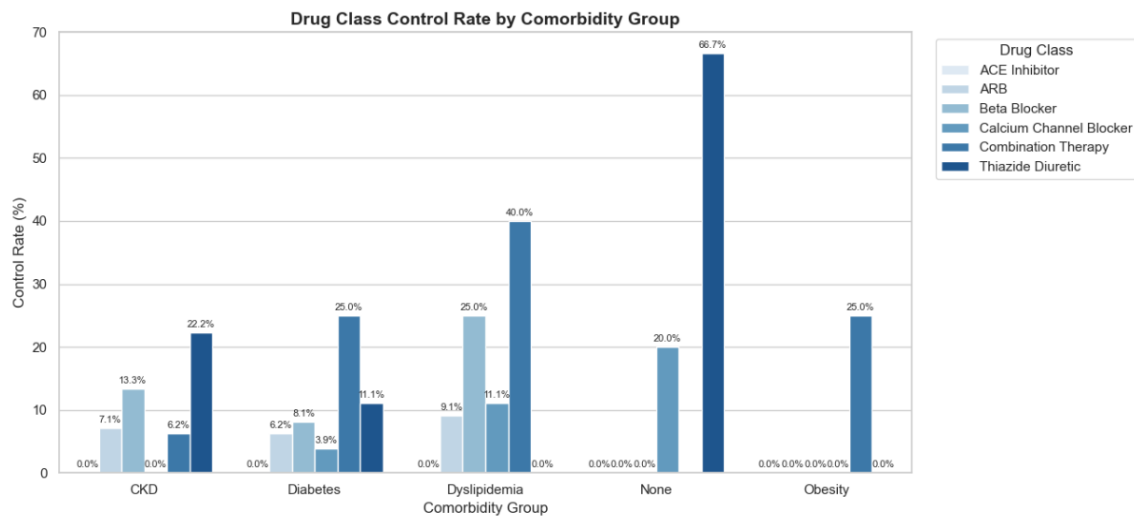
DRUG EFFECTIVENESS ACROSS COMORBIDITY PROFILES

This answers the second problem statement: Which antihypertensive drug class delivers the best outcomes for patients with specific comorbidity profiles?

For individual comorbidity:

- Chronic Kidney Disease:** Thiazide Diuretic works best with a control rate of 22.22%, followed by Beta Blocker(13.33%), ARB(7.14%), Combination Therapy(6.25%) with Ace Inhibitor and Calcium Channel Blocker has 0% control rate.

- **Diabetes Mellitus:** Combination Therapy works best with a control rate of 25% followed by Thiazide Diuretic(11.11%), Beta Blocker(8.11%), ARB(6.25%), Calcium Channel Blocker(3.85%) with Ace Inhibitor having 0% control rate.
- **Dyslipidemia:** Combination Therapy works best with control rate of 40% followed by Beta Blocker(25%), Calcium Channel Blocker(11.11%), ARB(9.09) with Thiazide Diuretic and ACE Inhibitor having 0% control rate.
- **Obesity:** Only Combination Therapy worked for Obese patients with 25% control rate.
- **None:** For patients with no comorbidity profile, Thiazide Diuretic works best with control rate of 66.67% followed by Calcium Channel Blocker(20%) while Beta Blocker, ARB, ACE Inhibitor have 0% control rate.



In conclusion, Combination Therapy delivers the most consistent and higher control rate after 3 months across different comorbidity profiles. Although Thiazide Diuretics, Calcium Channel Blocker, Beta Blocker and ARB have moderate but inconsistent performance with ACE Inhibitor having 0% control rate throughout. This means that patients with comorbidity profiles benefit more from combining several therapies than single-drug therapy.

For full comorbidity profiles, that is, patients with more than one comorbidity:

- Dyslipidemia & Obesity: Calcium Channel Blocker with 25% control rate.
- Chronic Kidney Disease and Obesity: Thiazide Diuretic with 25% control rate.
- Chronic Kidney Disease & Dyslipidemia: Beta Blocker with 20% control rate.
- Chronic Kidney Disease & dyslipidemia & Obesity: Thiazide Diuretic with 100% control rate.
- Diabetes Mellitus & Obesity: Beta Blocker with control rate of 50%.
- Diabetes Mellitus & Dyslipidemia: Beta Blocker with control rate of 50%.
- Diabetes Mellitus & Dyslipidemia & Obesity: Combination Therapy with control rate of 28.57%.

- Diabetes Mellitus and Chronic Kidney Disease: Thiazide Diuretic with control rate of 11.11%.
- Diabetes& Chronic Kidney Disease and Obesity: Combination Therapy with control rate of 33.33%.
- Diabetes & Chronic Kidney Disease and Dyslipidemia: Combination Therapy with control rate of 50%.
- Diabetes Mellitus & Chronic Kidney Disease & Dyslipidemia & Obesity: Thiazide Diuretic with a control rate of 28.57%.

PYTHON PREDICTIVE MODELLING

This answers the first problem statement: What predicts BP failure at 3 months, and can we identify high-risk patients at the point of prescription?

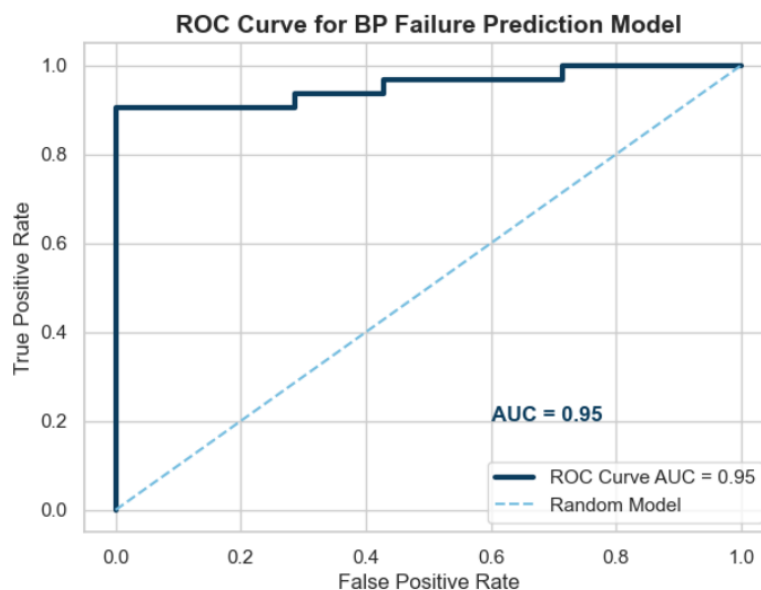
Procedure: I created a new target variable using BP_Failure = 1 if it was not controlled after 3 months and BP_Failure = 0 if it was controlled after 3 months.

Then I used features that are available at the point of prescription which includes: Age, Age group, Gender, Diabetes mellitus, Chronic Kidney Disease, Dyslipidemia, Obesity, Baseline Systolic BP, Baseline Diastolic BP, Antihypertensive Class, Medication Adherence and Number of Visits.

Model Performance

The model achieved:

- Accuracy: 92.86%
- Recall for BP Failure: 1.00
- ROC-AUC:0.95



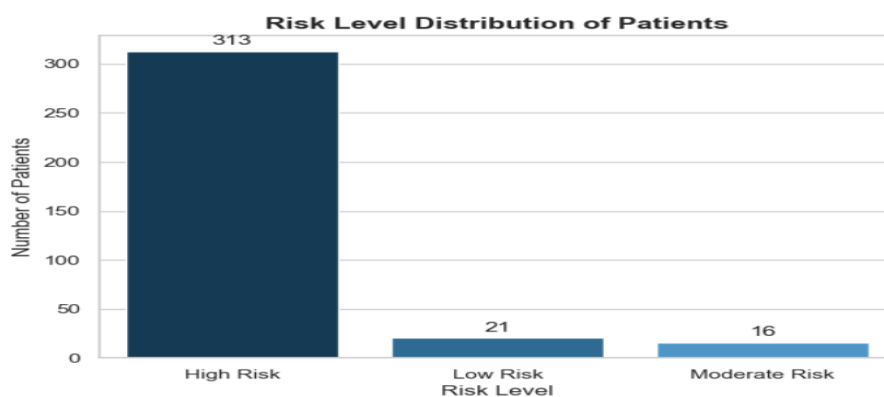
The ROC-AUC score of 0.95 indicates excellent model performance which means that the model has a strong ability to distinguish between patients who will achieve BP control and those who are likely to fail.

The recall score of 1.00 for BP failure means the model successfully identified all patients in the test set who failed BP control.

These are essential in healthcare as missing a high-risk patient will lead to poor outcomes.

The model is effective for identifying high-risk patients at the point of prescription. This supports early intervention and can help clinicians prioritize patients who need closer monitoring.

High- Risk Patient Identification



Using the predicted probability from the model, patients were grouped into risk levels:

- High Risk: Failure Probability $\geq 70\%$.
- Moderate Risk: Failure Probability between 40% and 69%
- Low Risk: Failure Probability below 40%

This makes the model practical as it does not only predict failure but also helps create a risk profile for each patient.

Clinicians can use this risk classification to identify patients who need more frequent visits, consider using combination therapy early and to monitor severe hypertension cases more closely.

KEY PREDICTORS OF BP CONTROL FAILURE

1. Baseline BP Severity.
2. Medication Adherence.
3. Comorbidities.
4. Antihypertensive Class.

RECOMMENDATIONS

1. **Treat severe baseline hypertension as high-risk:** Patients presenting with severe BP levels should receive closer monitoring from the point of prescription.
2. **Strengthen medication adherence programs:** Since adherence strongly influences BP outcomes, the hospital should introduce adherence counseling, reminders and patient education on the effects of not adhering to the medications given.
3. **Use personalized treatment strategies according to the comorbidity profile:** Drug effectiveness differs by comorbidity profile so patients should receive personalized treatment plan rather than a general treatment plan.
4. **Consider Combination Therapy for complex patients:** Combination Therapy shows more consistent and better performance across comorbidity groups and will be suitable for patients with multiple risk factors.
5. **Use predictive risk scoring:** The python model can be used to identify high risk patients early, allowing the clinician to intervene before treatment failure occurs.
6. **Improve follow-up care:** Patients with high predicted risk should be scheduled for closer follow-up visits and monitored regularly.

LIMITATION

The dataset provided has a small sample size limiting the interpretations to a small group so the hospital should be careful while using the results of the analysis as this limits generalization.

N.B: The predictive model should be validated with larger datasets before clinical deployment.

CONCLUSION

This project shows how healthcare data can be used to support better clinical decision-making and improve hospital performance.

This analysis revealed that baseline BP severity, medication adherence, comorbidities and drug class are important drivers of BP control outcome after 3 months. The predictive model achieved a ROC-AUC score of 0.95, showing excellent ability to distinguish high-risk patients from low-risk patients. With a recall score of 1.00 for BP failure, this model can be used to identify patients who will require early intervention.

For drug outcomes, combination therapy showed the most consistent performance across comorbidity profiles, suggesting that patients with complex health conditions may require more personalized treatment strategies.

This project demonstrates that data analytics can help hypertension centers move from reactive care to risk-based patient management.